

# ANGCHEN XIE

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## EDUCATION

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### Carnegie Mellon University

M.S. in Robotics (MSR)

Pittsburgh, PA

Aug. 2025 – May 2027 (expected)

Advised by [Guanya Shi](#) and [Max Simchowitz](#)

- GPA: 4.17/4.00
- Courses: Introduction to Robot Learning; Optimal Control and Reinforcement Learning; Computer Vision; Advanced Control for Robotics; Mechanics of Manipulation

### Shanghai Jiao Tong University

B.Eng. in Automation

Shanghai, China

Sept. 2021 – June 2025

- GPA: 4.01/4.30 | Rank: 1/87
- Undergraduate thesis: *Fault-Tolerant Control of Legged Robots under Multi-Sensor Fusion*

## RESEARCH INTERESTS

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I work toward general-purpose robots that combine broad learned priors with grounded models of themselves and the physical world, enabling reliable and versatile behavior across tasks, embodiments, and environments.

## PUBLICATIONS

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\* Equal contribution.

- [1] [Under review, CoRL 2026] **Angchen Xie\***, Nikhil Sobanbabu\*, Ishayu Shikhare, Alan Wang, Max Simchowitz, Guanya Shi. “FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control”.  
[📄 arXiv](#) [🌐 Website](#) [🐦 Thread](#)
- [2] [RSS 2026] Rishabh Madan, **Angchen Xie**, Samantha Saak, Andres Blanco, Dohyeok Lee, Sarah Grace Brown, Yunting Yan, Mark Zolotas, Jose Barreiros, Tapomayukh Bhattacharjee. “TACTIC: Tactile and Vision Conditioned Contact-Centric Control for Whole-Arm Manipulation”.  
[📄 arXiv](#) [🌐 Website](#)
- [3] [RSS 2026] Pablo Ortega-Kral\*, Eliot Xing\*, Arthur Buckner, Vernon Luk, Junseo Kim, Owen Kwon, **Angchen Xie**, Nikhil Sobanbabu, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussell, Guanya Shi, Jonathan Francis, Jean Oh. “RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning”.  
[📄 arXiv](#) [🌐 Website](#) [🔗 Code](#)
- [4] [CoRL 2025] Rishabh Madan, Jiawei Lin, Mahika Goel, **Angchen Xie**, Xiaoyu Liang, Marcus Lee, Justin Guo, Pranav N. Thakkar, Rohan Banerjee, Jose Barreiros, Kate Tsui, Tom Silver, Tapomayukh Bhattacharjee. “PrioriTouCh: Adapting to User Contact Preferences for Whole-Arm Physical Human-Robot Interaction”.  
[📄 arXiv](#) [🌐 Website](#)
- [5] [IROS 2024] **Angchen Xie**, Yeqiang Qian, Weihao Yan, Chunxiang Wang, Ming Yang. “Non-repetitive: A Promising LiDAR Scanning Pattern”.  
[📄 Paper](#) [📁 Dataset](#)

## RESEARCH EXPERIENCE

### LeCAR Lab, Carnegie Mellon University

Graduate Research Assistant

Pittsburgh, PA

Aug. 2025 – Present

Advised by [Guanya Shi](#) and [Max Simchowitz](#) | humanoid control and cross-embodiment infrastructure

- Proposed and developed a Planner-IDM framework that freezes the planner and finetunes only the IDM on two minutes of target-domain rollouts, adapting to changes in terrain, payload, and actuator dynamics without rewards or demonstrations and raising real-robot success from 20% to 90%. (*FADA*)
- Co-developed an open-source cross-embodiment framework for control, teleoperation, and policy deployment, reducing observation-to-action latency to 130 ms versus 581 ms with LeRobot; led its humanoid integration, including latency benchmarking, policy training, and hardware deployment. (*RIO*)

### RL<sup>2</sup> Lab, Shanghai Jiao Tong University

Undergraduate Research Assistant

Shanghai, China

Jan. 2025 – June 2025

Advised by [Yue Gao](#) | fault-tolerant locomotion for legged robots

- Developed a multi-sensor representation-learning controller using contrastive fault embeddings to adapt hexapod gait to joint failures, improving fault-diagnosis accuracy from 78.6% to 91.2% and maintaining locomotion under up to three simultaneous failures after Isaac Gym-to-MuJoCo transfer. (*Undergraduate thesis*)

### EmPRISE Lab, Cornell University

Research Intern

Ithaca, NY

July 2024 – Dec. 2024

Advised by [Tapomayukh Bhattacharjee](#) | contact-rich whole-arm physical human-robot interaction

- Developed the learned latent dynamics model, which fuses tactile force, RGB, and depth to drive a contact-centric whole-arm MPC controller, and built its simulation evaluation pipeline, reaching 79.5% success on simulated bed bathing versus 51.6% for DreamerV3 with 65% fewer force violations. (*TACTIC*)
- Built the Unity bed-bathing environment behind a learning-to-rank framework for conflicting force preferences, instrumenting the human model with force-sensor colliders and contributing to the ranking algorithm that significantly reduced force violations versus fixed and heuristic prioritization ( $p < 0.05$ ). (*PrioriTouch*)

### CyberC3 Intelligent Vehicle Lab, Shanghai Jiao Tong University

Undergraduate Research Assistant

Shanghai, China

Nov. 2022 – June 2024

Advised by [Yeqiang Qian](#) and [Ming Yang](#) | LiDAR perception for intelligent vehicles

- Built *Repetitive-or-not*, a CARLA benchmark isolating the LiDAR scanning pattern as the only variable, and showed that non-repetitive scanning captures up to 59.6% more traffic participants, with larger gains for smaller targets such as pedestrians, while quantifying the domain gap in 3D object detection. (*Non-repetitive*)

## SERVICE

Reviewer, Conference on Robot Learning (CoRL)

2026

Reviewer, Robotics: Science and Systems (RSS)

2026

Reviewer, IEEE Robotics and Automation Letters (RA-L)

2025, 2026

## SKILLS

### Programming

Python, C/C++, MATLAB,  $\LaTeX$

### Frameworks & Simulators

PyTorch, ROS/ROS2, Isaac Gym, Isaac Lab, MuJoCo, Gazebo, CARLA, Unity

### Robot Platforms

Unitree G1, Unitree Go1/Go2, AgiBot G1, Booster T1, Kinova

## HONORS AND AWARDS

Outstanding Graduate Award, Shanghai Jiao Tong University

2025

China National Scholarship (top 0.2%)

2024

First Prize Academic Scholarship, Shanghai Jiao Tong University

2024

National First Prize, China Undergraduate Electronics Design Contest (top 1%)

2023

First Prize, China Undergraduate Mathematical Contest in Modeling (top 2%)

2022